

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	2 July 2026
Team ID	24A55A0518
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

Technical Architecture:

TECHNICAL ARCHITECTURE – METRO TICKET GENERATING SYSTEM

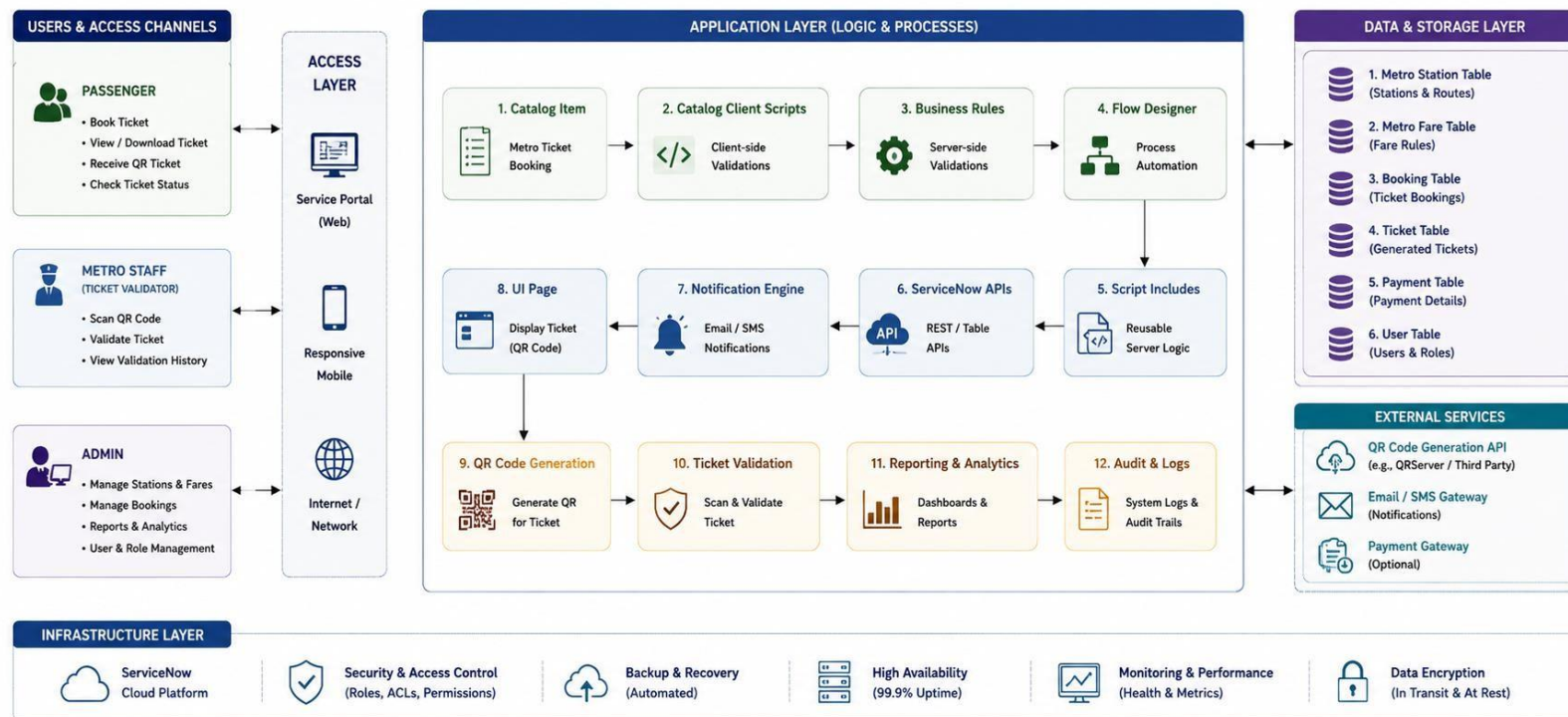


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides a web-based interface through Service Portal, Catalog Items, and UI Pages for passengers to book tickets and view QR tickets	ServiceNow Service Portal, Catalog Items, UI Pages
2.	Application Logic-1	Performs client-side validations, fare calculations, and dynamic form operations	Catalog Client Scripts (JavaScript)
3.	Application Logic-2	Handles server-side processing such as ticket generation, validation, and database operations	Business Rules (Server-side JavaScript)
4.	Application Logic-3	Automates booking workflow, notifications, and ticket processing	Flow Designer, Script Include
5.	Database	Stores ticket details, passenger information, booking records, and validation status	ServiceNow Custom Tables (Metro Database, Metro Station)
6.	Cloud Database	Provides cloud-based storage and data management services	ServiceNow Cloud Database
7.	File Storage	Stores QR code images, attachments, and generated ticket data	ServiceNow Attachment Service
8.	External API-1	Generates QR codes for digital metro tickets	QRServer API (Third-party QR Generation API)
9.	External API-2	Facilitates platform integration and data exchange	ServiceNow REST API
10.	Machine Learning Model	No machine learning model is implemented in this project	Not Applicable
11.	Infrastructure (Server / Cloud)	Hosts and deploys the Metro Ticket Generating System	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	JavaScript libraries and ServiceNow framework	JavaScript, ServiceNow
2.	Security Implementations	ACLs, Roles, Access Controls	ServiceNow ACL, RBAC
3.	Scalable Architecture	Cloud-based modular architecture	ServiceNow Cloud
4.	Availability	24×7 cloud availability	ServiceNow SaaS
5.	Performance	Real-time fare calculation and QR generation	ServiceNow Platform

References:

<https://c4model.com>

<https://goqr.me/api>

https://www.servicenow.com/docs/bundle/latest-build-workspace/page/script/server-scripting/concept/c_ScriptIncludes.html

<https://www.servicenow.com/docs/bundle/latest-build-workspace/page/administer/flow-designer/concept/flow-designer.html>

<https://www.servicenow.com/docs/bundle/latest-build-workspace/page/build/service-portal/concept/service-portal.html>